Goal: How well does the inverse + surrogate pipeline generalize to Quantum Metal parameters it has never seen before

Probe the in-between regions of SQuADDS by sampling uniformly inside the training cloud and binning by proximity.

1. Sample uniformly

Generate 50,000 uniformly random Quantum Metal parameter sets inside the convex hull of the training data (pure interpolation, no extrapolation).

2. Measure NN distance

For each random point, compute d_{NN} = Euclidean distance to its nearest training sample in the scaled $[0, 1]$ parameter space.

3. Bin and select

Split the d_{NN} range into 10 equal-width bins (close $\rightarrow$ far). Randomly draw 9 points per bin $\rightarrow$ 90 total for Ansys validation.

Surrogate evaluates all 50,000 samples in seconds.
90 selected across distance bins sent to Ansys for validation.